JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 91, NO. B11, PAGES 11,545-11,555, OCTOBER 10, 1986

PALEOMAGNETISM OF JURASSIC VOLCANIC ROCKS IN THE PATAGONIA MOUNTAINS, SOUTHEASTERN ARIZONA: IMPLICATIONS FOR THE NORTH AMERICAN 170 MA REFERENCE POLE

Steven R. May[1], Robert F. Butler, Muhammad Shafiqullah, and Paul E. Damon

Department of Geosciences, University of Arizona, Tucson

Abstract. Paleomagnetic study of Middle Jurassic volcanic and volcaniclastic rocks in southeastern Arizona yields a paleomagnetic pole located at 61.8°N, 116.0°E (k=49.6, α_{95}=6.2°). These rocks from Corral Canyon represent an autochthonous segment of the Jurassic Cordilleran magmatic arc and were deposited directly on Upper Paleozoic carbonates of the cratonic miogeocline. Rock magnetic experiments demonstrate that the natural remanent magnetization of these rocks is dominantly a single component magnetization carried in both magnetite and hematite phases and is interpreted as a primary thermoremanent magnetism. Rb/Sr isotope geochemistry on whole-rock samples from eight welded ash flow tuffs yields an isochron age of 172.2±5.8(2σ) Ma. The Corral Canyon pole falls between poles from Newark Trend Group II (179±3 Ma) and Glance Conglomerate (151±2 Ma).

Introduction

Our understanding of the Jurassic apparent polar wander (APW) path for North America has been hampered by the paucity of well dated Jurassic rocks on the craton whose paleomagnetic record can be unambiguously interpreted. Including a new paleomagnetic pole (paleopole) reported in this study, there are only eight reliable paleopoles representing some 70 m.y. of North American Jurassic plate motion [May and Butler, this issue]. Yet from a global tectonic perspective, this period of polar wander history is critical, as it records the breakup of Pangea and the separation of North America from the southern continents. Paleomagnetic poles provide the primary data with which geologists can reconstruct global and regional paleogeographies within an absolute paleolatitudinal context. In addition, cratonic reference paleopoles are important to documentation of latitudinal displacements and azimuthal rotations of terranes within the western Cordillera [Hillhouse, 1977; Beck, 1980].

As reviewed by Steiner [1983] and Gordon et al. [1984], much of what we know about Jurassic APW is based on paleomagnetic studies from sedimentary rocks exposed on the Colorado Plateau. Systematic discrepancies between paleopoles of various ages suggest that the Colorado Plateau may have undergone approximately 4±3° of clockwise rotation relative to rocks east of the Rocky Mountains in post-Jurassic time [Bryan and Gordon, 1985]. Whether or not this model proves correct, it does accentuate the need for Jurassic paleomagnetic poles from other parts of cratonic North America.

The geology of southeastern Arizona provides an opportunity to study the paleomagnetism of Middle and Late Jurassic volcanic rocks that were erupted onto a Precambrian and Paleozoic cratonic basement. These volcanic rocks and associated granitic plutons, hypabyssal intrusives, and volcaniclastic sediments represent an autochthonous segment of the Cordilleran magmatic arc associated with east dipping subduction along the western continental margin [Coney, 1978]. Other segments of this magmatic arc in western Canada and Alaska exist as suspect terranes whose paleogeographic relationship with respect to the North American craton is uncertain [Coney et al., 1980]. The Jurassic arc is also known to extend south into Mexico [Damon et al., 1981]; however, structural complications associated with Mesozoic strike-slip faults (e.g., Mojave-Sonora Megashear) preclude the unambiguous interpretation of potential paleomagnetic data from Mexico in terms of North American APW.

Recent geologic work [Kluth, 1982, 1983; Vedder, 1984] has elucidated certain stratigraphic and structural relationships within the Jurassic arc assemblages of southeastern Arizona, thereby much facilitating paleomagnetic investigations. Kluth et al. [1982] have reported a Late Jurassic paleopole from volcanic rocks in the Canelo Hills, and we present here the results of a paleomagnetic-geochronologic-stratigraphic study of slightly older Jurassic rocks in the Patagonia Mountains. Results include a well-dated paleomagnetic pole, from a sequence of interbedded ash flow tuffs and nonmarine sediment, that is consistent both spatially and temporally with radiometrically dated poles from southern Arizona [Kluth et al., 1982] and the eastern United States [Smith and Noltimier, 1979].

The geologic time scale used in this paper is that of Harland et al. [1982], which places the Jurassic-Cretaceous boundary at 144 Ma and the Jurassic-Triassic boundary at 213 Ma.

Regional Geology

The Patagonia Mountains are a north-northwest trending range located in the southern Basin and Range province northeast of Nogales, Arizona (Figure 1). The geology of the Patagonia Mountains has been discussed by Stoyanow [1949], Hayes [1970], Simons [1972], and Hayes and Drewes [1978] and mapped by Simons [1974]. The core of the Patagonia Mountains is composed of an early Tertiary granodioritic pluton with Precambrian through late Mesozoic plutonic and supracrustal rocks exposed on both east and west flanks.

Along the eastern side of the Patagonia Mountains, Permian Concha Limestone is overlain unconformably by silicic volcanic rocks of Middle Jurassic age. This volcanic sequence, which is

[1]Now at Exxon Production Research Company, Houston, Texas.

Copyright 1986 by the American Geophysical Union.

Paper number 6B5933.
0148-0227/86/006B-5933$05.00

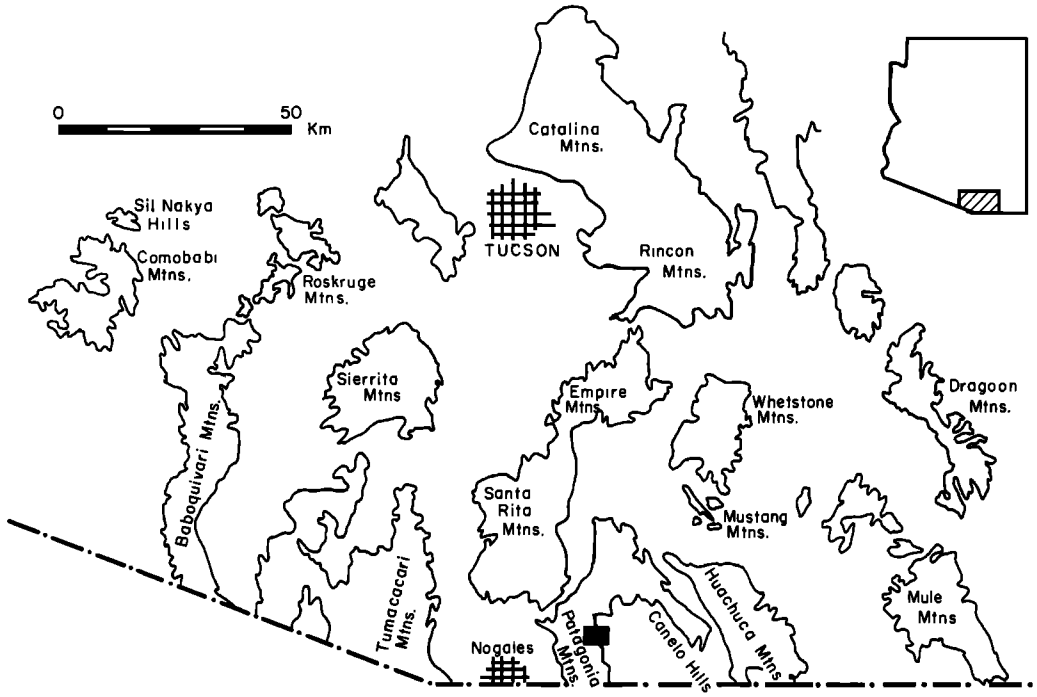


Fig. 1. Index map of southeastern Arizona showing the location of the Patagonia Mountains. Black square is area of Figure 2.

informally referred to as the Duquesne volcanics [Baker, 1961], includes silicic lavas, welded tuffs, flow breccias, and thin quartzite beds and may be as much as 1-2 km thick [Simons, 1972]. North of Corral Canyon, the Duquesne volcanics overlie the thick sequence of interbedded ash flow tuff, andesitic lava, and volcaniclastic sediment, which were the target of this study (Figure 2). West of the Harshaw Creek fault, the Duquesne volcanics are overlain unconformably by Lower Cretaceous clastic sediments of the undifferentiated Bisbee Formation [Hayes, 1970]. Lithologic similarities and relative stratigraphic position suggest correlation of the Duquesne volcanics with the Canelo Hills volcanics (as restricted by Kluth [1982] and Vedder [1984]). The Canelo Hills volcanics have been radiometrically dated (K-Ar) at 169±6 and 177±8 Ma [Marvin et al., 1978].

Rocks of Corral Canyon

A 650-m-thick sequence of red mudstone, sandstone, conglomerate, ash flow tuff, and andesitic lava is exposed north of the Corral Canyon fault (Figure 2). These rocks were mapped by Simons [1974] as "JTrvs" and described by Simons [1972] as the "volcaniclastic sequence" of the Corral Canyon block. The name "Corral Canyon Formation" has been used in various unpublished theses [Baker, 1961; Knight, 1970; Kluth, 1982]; however, an informal nomenclature is retained in this paper.

The Corral Canyon rocks form an east-west striking, north facing faulted homocline with dips ranging from 26° to 68° (Figure 2). Their southern boundary is the Corral Canyon fault [Simons, 1972], an apparently steep normal fault, which removes an unknown thickness of section. As much as 100 to 150 m of Corral Canyon strata strike into the fault as one moves upsection from southwest to northeast. The western boundary of the Corral Canyon outcrop area is the American fault, which records at least hundreds of meters of Jurassic vertical displacement. To the north, the Corral Canyon rocks are depositionally overlain by the Duquesne volcanics; the contact between these units is arbitrarily placed at the base of an easily recognizable and widespread welded ash flow tuff.

The Corral Canyon section can be subdivided into three informal members for the purpose of description. The lower member has a maximum exposed thickness of 330 m and consists of approximately 50% sediments and 50% ash flow tuff and tuff breccia. The lowest exposed strata consist primarily of red mudstone and sandy mudstone as well as volcanic lithic sandstone, conglomerate, chert beds, and thin ash flow tuffs. The lower member sediments are interpreted as distal(?) alluvial fan deposits. Fining upward sandstone and mudstone sequences represent shallow channel and overbank deposits while thin, matrix-rich sandstones may represent sheet flood deposits.

Overlying and interfingering with the red beds of the lower member is an assemblage of volcanic strata, which includes purple, gray, and white welded ash flow tuff, greenish brown andesite, and light purple-gray volcanic flow breccia. In field appearance, the ash flow tuffs are heterogeneous and display well-developed eutaxitic texture and lithic inclusions. Flow thicknesses range from 1 to 10 m and are quite variable along strike. Phenocrysts of plagioclase, sanidine, quartz, and biotite are common in a devitrified glassy groundmass.

A 50- to 100-m-thick interval of variably

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11545 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

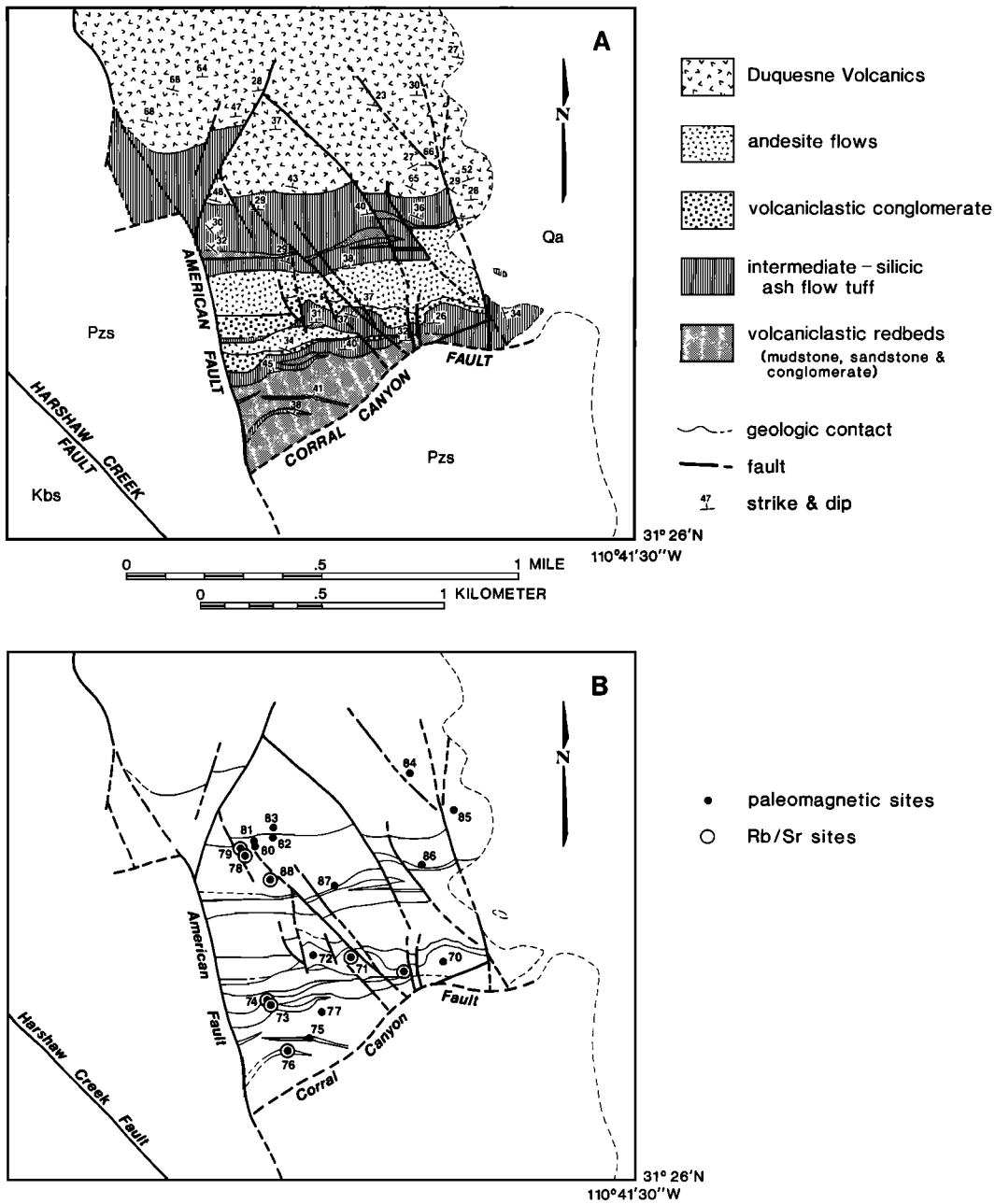


Fig. 2. (a) Geologic map of the Corral Canyon area. Qa, Quaternary alluvium; Kbs, Cretaceous Bisbee Group sediments; and Pzs, Paleozoic sediments. (b) Locations of paleomagnetic and Rb/Sr sample sites. Site numbers are keyed to paleomagnetic sites listed in Table 2 (e.g., 83 = CH083).

altered andesitic lavas comprises the middle member of the Corral Canyon sequence (Figure 2). These lavas are greenish brown to green in color and exhibit only a crude stratification, which is obscured by the weathered state of most outcrops. No samples for either paleomagnetic or Rb/Sr analysis were collected from these andesites.

Overlying the andesite flows is an upper member, which includes gray, pink, and purple welded ash flow tuffs with subordinate lavas and volcaniclastic sediments. Individual tuffs vary from a few meters to a few tens of meters in thickness, while the thickness of the upper member changes dramatically across the Corral Canyon outcrop area from east to west (Figure 2). The observed geometry of stratigraphy and structures is interpreted to reflect growth of a NNW-SSE oriented graben system contemporaneous with, and controlling deposition within, the Corral Canyon sequence.

The upper part of the Corral Canyon stratigraphy is exclusively volcanic in composition, resulting in an arbitrarily placed formational contact with the overlying Duquesne volcanics at the base of an easily recognized purple, densely welded ash flow tuff. Two of the three paleomagnetic sites within the Duquesne volcanics yield reliable mean directions indistinguishable from

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11545 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Rb/Sr Isotopic Date From the Corral Canyon Sequence

Sample	Rb, ppm	Sr, ppm	$^{87}Rb/^{86}Sr$	$^{87}Sr/^{86}Sr$
UARS 84-150	182.5	132.1	3.990	0.71860±0.00002
UARS 84-151	303.9	92.8	9.496	0.73164±0.00002
UARS 84-153	82.8	351.9	0.681	0.71038±0.00003
UARS 84-1	97.9	299.6	0.946	0.71058±0.00003
UARS 84-20	50.0	355.2	0.408	0.70952±0.00002
UARS 84-21	112.4	297.5	1.094	0.71116±0.00002
UARS 84-130	107.4	45.6	6.826	0.72511±0.00002
UARS 84-133	121.1	192.5	1.821	0.71254±0.00003

site mean directions of the Corral Canyon rocks, suggesting that these two assemblages are not separated by a significant interval of time. Further description of stratigraphy and structures can be found by May [1985].

Rb/Sr Geochronology

Eight whole-rock samples from the Corral Canyon ash flow tuffs were analyzed for Rb and Sr isotopes to determine an isochron age (Table 1, Figure 2). Strontium was separated by conventional ion exchange techniques and analyzed using a VG 354 mass spectrometer in the triple collector mode. Total strontium and rubidium were determined by isotope dilution. Analysis of the NBS 987 interlaboratory standard $SrCO_3$ gave a mean of 0.71025. All isotopic analyses were normalized to an $^{86}Sr/^{88}Sr$ ratio of 0.1194. The mean experimental uncertainties were taken as ±1% for $^{87}Rb/^{86}Sr$ for the data listed in Table 1. Care was taken during sample preparation to remove any fragments in which lithic fragments were observed.

The eight samples can be separated into two groups based on stratigraphic sequence. The model ages are in correct order, 172 Ma for the lower group and 170 Ma for the upper group, but the difference is not statistically significant. Accordingly, the two groups have been combined into one isochron (Figure 3), which yields an age of 172.2±5.8(2σ) Ma using the model 3 solution of Ludwig [1983]. This model assumes that the scatter in excess of the estimated analytical error is a result of gaussian distributed variation in the initial ratio of the samples. The initial ratio for the isochron is 0.7085±0.0003(2σ). Although high, the initial ratio is within the range of 0.7069 to 0.7096 observed for mid-Tertiary volcanic rocks in southeastern Arizona [Shafiqullah et al., 1978] and is also similar to the 0.7092 value reported by Kluth et al. [1982] for Late Jurassic ash flow tuffs in the neighboring Canelo Hills.

Consistency of the 172.2±5.8 Ma age with K-Ar ages obtained from the Canelo Hills volcanics, the unmetamorphosed nature of the interbedded sedimentary rocks, and the paleomagnetic data indicating recording of geomagnetic secular variation during extrusion of the Corral Canyon sequence indicate that this Rb/Sr isochron age dates the extrusion of the Corral Canyon volcanic rocks and not a later metamorphic event.

Paleomagnetism

There were 126 cores collected from 19 sites in the Corral Canyon sequence (Figure 2). Standard paleomagnetic coring techniques were used, and core orientations were determined with a magnetic compass. None of the rocks sampled were sufficiently strongly magnetized to require sun compass orientation. This was verified by sun compass orientation of coring holes at one of the lightning-struck sites (CH086, see below), where intensities of magnetization were much higher than usual for this volcanic sequence. Sun compass orientations were within 1° of those determined with magnetic compass.

The 650 m of stratigraphic coverage included 18 sites that were collected in separate ash flow tuffs and one site (CH077) from a well-indurated red mudstone. Commonly, a single sample was cut from each core except in cases where two specimens could be obtained to analyze within-core variation. The structural attitude of each tuff was determined by measuring the orientation of flattened pumice lapilli. Such measurements were consistent with the attitudes of nearby sedimentary strata. Structural correction was made by simple rotation about strike.

The natural remanent magnetism (NRM) of all samples was measured with an ScT two-axis cryogenic magnetometer. Alternating field (AF) demagnetization was performed with a Schonstedt GSD-1 demagnetizer equipped with a tumbler, while thermal demagnetization was done in a mu-metal

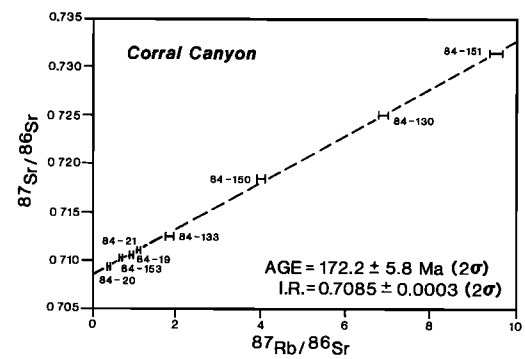


Fig. 3. Rb/Sr isochron for all eight analyses of whole rock tuff samples from the Corral Canyon sequence. Sample numbers next to data points are keyed to Table 1.

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11545 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

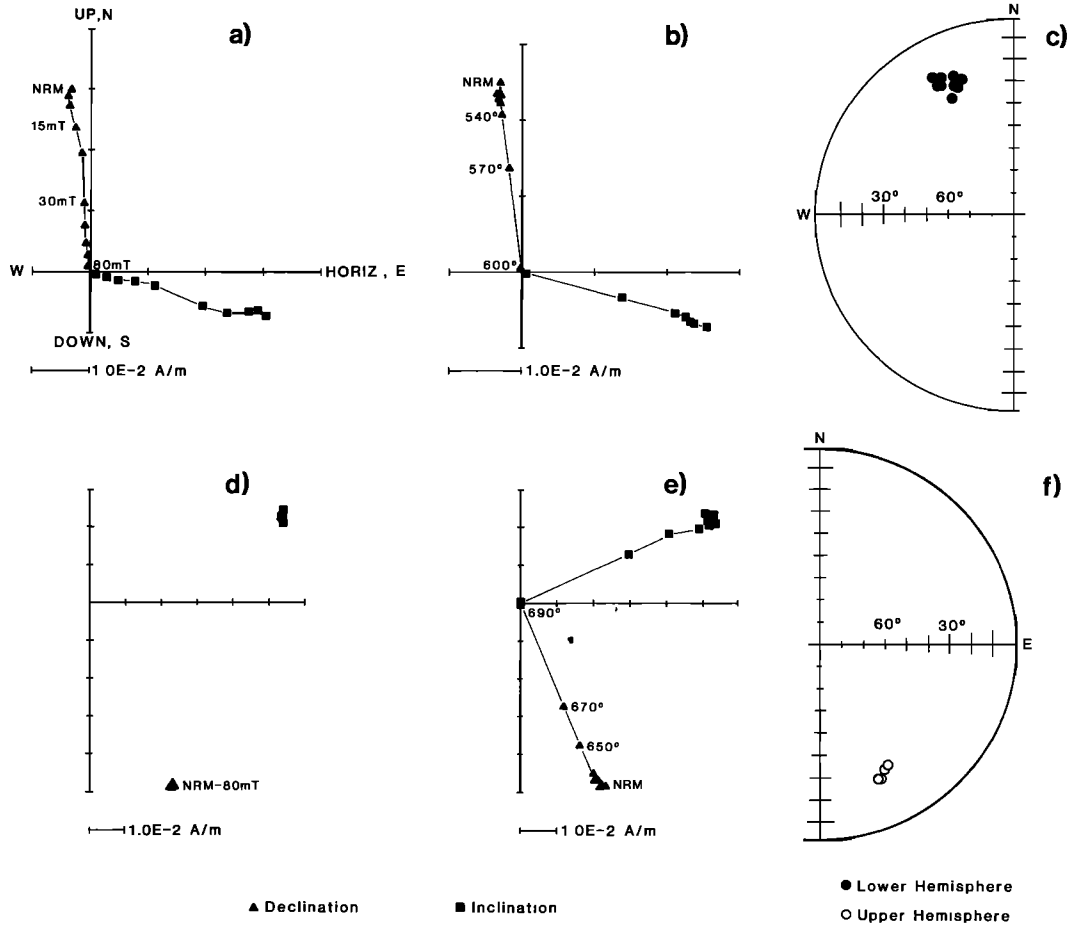


Fig. 4. Vector demagnetization diagrams and equal-area projections of cleaned, structurally corrected sample directions for (a, b, and c) site CH072 and (d, e, and f) site CH080. The orientation of axes for all examples is as shown in Figure 4a. Alternating field (mT) and thermal demagnetization (°C) steps are shown next to representative points.

shielded furnace with an ambient field of less than 15 nT (1 nT = 1γ).

Analysis of Demagnetization Experiments

Initial NRM intensities ranged from 10^{-1} to 10^{-3} A/m (1 A/m = 10^{-3} emu/cm^3), and within-site magnetic directions were generally well grouped except for a few cases in which lightning-induced components produced significant scatter. The effect of lightning strikes varied from producing one or two outliers within a site to a near randomization of sample directions. Coincident with the occurrence of anomalous directions was the observation of initial NRM intensities higher than the norm but still always less than 10^{-1} A/m. Pervasive lightning-strike remagnetization resulted in deletion of all data from three of the original 19 sites.

Both alternating field and thermal demagnetization experiments were performed to evaluate the directional stability and coercivity/blocking temperature spectra of each site. Vector demagnetization diagrams were used in this analysis. At least one sample from each site was progressively demagnetized in six to 10 steps at peak fields ranging from 5 to 80 mT (1 mT = 10 Oe). Two samples were commonly treated for sites that possessed outliers of suspected lightning-strike origin. Such outliers commonly experienced rapid NRM intensity loss in peak fields of less than 20 mT and often moved toward the site mean direction. During progressive AF demagnetization, a spectrum of behavior was observed, the endmembers of which are illustrated in Figure 4.

Samples from six sites responded to AF demagnetization such that at least 50% of their initial intensity was removed after exposure to a peak field of 80 mT (Figure 4a). A minor secondary component was removed at low AF steps (<20 mT), after which a single, characteristic component was isolated. Progressive thermal demagnetization of pilot samples from these same six sites produced similar behavior with linear decay of the NRM toward the origin in vector demagnetization diagrams. For samples such as CH072B and D (Figures 4a and 4b) with coercivities less than 80 mT, the NRM is completely unblocked by 600°C. Samples that still retain a significant fraction of their initial intensity after

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11545 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

exposure to 80 mT peak demagnetizing fields also exhibit unblocking temperatures higher than 600°C.

In order to quantitatively confirm the single component, characteristic nature of the NRM isolated by AF demagnetization to 20 mT in these six sites responding to AF demagnetization, we subjected the progressive demagnetization results to component analysis by the least squares line technique of Kirschvink [1980]. Lines were fit to the progressive demagnetization data using three options: (1) fitting only the measured partially demagnetized NRM data alone, (2) fitting the demagnetization data and adding the origin as an additional, equally weighted point, and (3) anchoring the least squares line to the origin of the vector demagnetization plot.

Results for sample CH072B (Figure 4a) are typical of AF demagnetization behavior for which the line-fitting technique was applied to the demagnetization interval 20-80 mT. Directions resulting from the three line-fitting options are all within 1.5°, less than the mean angular deviation of fitting the lines to the data. In addition, no individual NRM direction in the 20-80 mT interval deviated from any of the line-fitting results by more than 5.1°. These results clearly demonstrate that the direction of NRM isolated by AF demagnetization in the 20-80 mT interval is both single component and a characteristic direction in the sense that the trend of the progressive demagnetization data includes the origin of the vector demagnetization diagram.

Progressive thermal demagnetization data for sample CH072D (Figure 4b) also indicate that the NRM unblocked by thermal demagnetization in the 490°-600°C interval is a single-component, characteristic NRM. In this case, all line-fitting options yield directions within 0.6°, and the maximum deviation of any individual NRM direction in the 490°-600°C interval from the line fitting results is 1.5°.

Pilot samples from seven sites exhibit only minor response to AF demagnetization, retaining 70-90% of the initial remanent magnetization after exposure to a peak alternating field of 80-100 mT. Thermal demagnetization of pilot samples from these sites yields simple, single-component vector demagnetization diagrams with unblocking temperatures above 600°C. Three sites exhibit no response to AF demagnetization but show excellent single-component decay during thermal demagnetization. Unblocking temperatures are distributed above 625°C with as much as 50% of the initial intensity blocked above 670°C (Figures 4d and 4e). Line-fitting analysis again confirms the single-component, characteristic nature of the NRM unblocked between 625° and 690°C. Directions resulting from the three line-fitting options applied to the progressive thermal demagnetization data of sample CH080E (Figure 4e) are within 1.2°. The maximum deviation between an individual NRM direction in the 625°-690°C interval and the line-fitting results is 1.5°.

Demagnetization experiments suggest a simple relationship between coercivity (H_c) spectra and unblocking temperatures (T_b) for the Corral Canyon rocks. The range of behavior includes sites that have total coercivity less than 80 mT and unblocking temperatures below 600°C to sites that have total coercivity above 80-100 mT and unblocking temperatures between 600° and 690°C. The former case (low H_c, low T_b) is interpreted as magnetite behavior, and the latter (high H_c, high T_b) as hematite. Intermediate behavior is indicative of a hybrid magnetic population containing both magnetite and hematite in variable proportions. Highly oxidized sites (i.e., those with high coercivities and blocking temperatures distributed above 600°C) also tended to have higher initial NRM intensities.

Isothermal remanent magnetization (IRM) acquisition experiments were conducted by magnetizing selected samples from eight representative sites in progressively higher magnetic fields up to 0.8 T (8000 Oe) and measuring the resultant remanent magnetization. A range of observed IRM curves is consistent with AF and thermal demagnetization experiments on samples from the same sites. As predicted for hematite [Dunlop, 1972], samples from sites with high coercivity and high blocking temperatures fail to reach saturation. IRM curves for samples from sites with low coercivity and blocking temperatures less than 600°C reveal significantly smaller proportions of IRM acquired in magnetic fields between 0.3 and 0.6 T. However, these samples also fail to reach total saturation, suggesting that while magnetite may be the dominant ferromagnetic phase, small amounts of hematite are probably also present.

The recognition of both magnetite and hematite phases within subaerially erupted volcanic rocks has been reported in a number of previous studies, including those by Kluth et al. [1982] and Hillhouse and Gromme [1984]. This association is commonly attributed to high-temperature (deuteric) oxidation of early crystallized magnetite grains either within a magma chamber or during eruption and emplacement. The degree to which hematite replaces magnetite in this process is a function of temperature and oxygen-pressure conditions [O'Reilly, 1984]. A deuteric oxidation model is strongly implicated in the magnetization history of the Corral Canyon ash flow tuffs. Statistical analysis of vectors removed during progressive AF and thermal demagnetization on individual samples demonstrates that both magnetic phases record the same magnetization direction within 95% confidence limits. We thus interpret the magnetic properties data to indicate that the NRM is a combination of thermoremanent magnetization (TRM) carried in magnetite grains and thermochemical remanent magnetization (TCRM) acquired during high temperature (deuteric) oxidation of magnetite to hematite. The relative importance of these two magnetization processes varies between ash flows and was dependent on temperature and composition during eruption and cooling.

Paleomagnetic Results

Final demagnetization treatment for all samples from each site was individually prescribed on the basis of the pilot demagnetization experiments. Multiple AF steps were used to isolate the primary direction in sites that responded well to AF demagnetization. Both AF and thermal demagnetization were used on sites that exhibited both magnetite and hematite properties and only thermal demagnetization was used for the three sites that showed no response

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11545 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. Site Paleomagnetic Data From the Corral Canyon Sequence

Site	N	Demag.	Dec., deg.	Inc., deg.	α_{95} deg.	k	R	J, A/m 10^{-3}A/m
CH070	6	H60	353.99	15.46	5.2	164.9	5.97	6.3
			348.30	46.18				
CH071	8	T650	332.26	23.93	3.5	250.4	7.97	120
			320.06	40.30				
CH072	8	H60	353.60	18.91	6.1	84.0	7.92	4.8
			347.96	47.80				
CH073*	8	T650	303.01	17.80	4.2	176.5	7.96	30
			270.57	50.81				
CH074*	8	T670	296.02	31.83	3.5	251.8	7.97	250
			252.61	51.60				
CH075	6	H40	327.96	14.50	11.0	38.1	5.87	3.3
			310.34	36.65				
CH076	7	H60	325.51	-2.90	3.9	237.1	6.97	1.2
			320.47	21.58				
CH077	8	H30	338.72	23.51	7.4	56.7	7.88	3.1
			317.91	58.61				
CH078*	6	T655	295.71	-38.93	9.5	50.7	5.90	5.6
			308.14	0.01				
CH080	5	T655	153.67	-24.95	2.8	769.7	4.99	40
			124.46	-52.52				
CH081	6	T610	354.43	27.33	14.8	21.4	5.77	2.5
			333.11	65.63				
CH082	3	T550	340.09	17.99	17.4	51.2	2.96	1.7
			333.37	55.10				
CH083	6	T650	343.62	40.44	5.2	165.6	5.97	38
			317.26	78.78				
CH084*	5	H60	125.65	-72.76	3.2	564.0	4.99	30
			26.15	-62.98				
CH085	3	T490	334.45	24.29	9.9	154.6	2.99	1.0
			328.97	51.49				
CH087	5	H40	337.02	7.24	13.3	34.1	4.88	2.6
			330.26	48.71				

Structurally corrected data for each site given first, followed by uncorrected direction. N, number of samples; Demag., demagnetization treatment (AF demagnetizations are labeled H where number indicates peak field in mT, thermal demagnetizations labeled T where number indicates temperature in °C); Dec., site mean declination; Inc., site mean inclination; α_{95}, radius of cone of 95% confidence about mean direction; k, Fisher precision parameter; R, resultant vector length; J, site mean intensity of remanent magnetism after cleaning.

*Sites excluded from formation mean.

to AF treatment. All samples whose NRM directions and intensities suggested the presence of lightning induced magnetizations were progressively demagnetized in alternating fields before further treatment.

Vector demagnetization diagrams were used both to evaluate the stability of each sample and, along with statistical parameters of Fisher [1953], to facilitate the selection of optimum demagnetization steps for site mean averaging. Results of the progressive demagnetization experiments described above clearly indicate the single-component, characteristic nature of the NRM isolated by the AF and thermal demagnetizations. Because of the unusually uncomplicated nature of the NRM, final site mean directions of NRM are insensitive to the selection of the particular demagnetization treatment. Any demagnetization treatment within the interval where the characteristic component is being removed would yield the same site mean direction. Equal-area projections of thermally cleaned sample directions from two representative sites are shown in Figures 4c and 4f.

All samples from sites CH079, CH086, and CH088 were rejected because of failure to isolate a stable direction. Initial intensities for samples in these sites were very high, and NRM directions were poorly grouped. The magnetization of these samples was so severely perturbed by lightning-induced IRM that the primary component could not be recovered. Individual samples from several other sites were also rejected due to instability during demagnetization or inadequate removal of anomalous directions.

Fisher [1953] statistics were applied to a final data set of 98 samples from 16 sites to calculate site mean directions and confidence parameters. These data are listed in Table 2 and illustrated in Figure 5. Thirteen sites are of normal polarity, while only one shows unambiguous reversed polarity. Two of the 16 sites (CH078

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11545 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

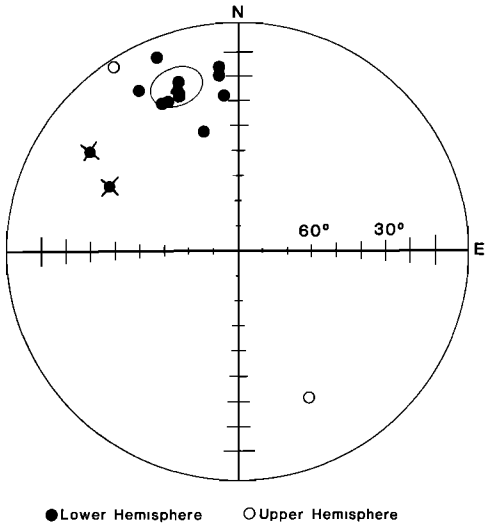


Fig. 5. Equal-area projection of selected, cleaned, structurally corrected site mean directions with formation mean (triangle) and α_{95} confidence circle. Site means with crosses are excluded from calculation of formation mean direction for reasons discussed in text.

and CH084) have very anomalous, although well-determined, directions and are interpreted as having recorded geomagnetic field polarity transitions or excursions. Sites CH078 and CH084 are not included in the calculation of a formation mean direction because of their clear discordance with respect to the main cluster of site mean directions.

Sites CH073 and CH074 also appear as outliers with respect to the main grouping of site mean directions. Rather than representing transitional field directions, the westerly declination of these sites is considered to reflect local vertical axis rotation associated with strike-slip motion along the American fault. Sites CH073 and CH074 are quite close to this fault, and their structural attitude is anomalous for this area of the Corral Canyon block. Rather than attempt to correct these structurally disturbed sites, we have decided to exclude them from the formation mean calculation.

The single reversed polarity site (CH080) is antipodal to the mean of the normal polarity sites (normal mean: declination (dec.), 340.0°; inclination (inc.), 19.4°; α_{95} = 8.5°; inverted CH080 dec, 333.7°, inc, 25.0°, α_{95} = 2.8°). Variation in bedding attitude is not sufficient to provide a definitive fold test, but the Fisher precision parameter (k) does increase from 20.3 to 40.96 in virtual geomagnetic pole (VGP) space and from 23.5 to 32.2 in direction space after structural corrections are applied to in situ site mean data. The better clustering of site directions after tilt correction is statistically significant at the 75% confidence level and indicates that magnetization was acquired prior to folding, which was probably a Late Cretaceous (Laramide) event. In addition, the k value for the formation mean direction indicates that geomagnetic secular variation has been adequately sampled.

Uncomplicated demagnetization behavior, well determined site mean directions spanning 650 m of stratigraphic section, a positive reversal test, and dispersion of the site mean directions indicative of sampling of geomagnetic secular variation suggest that the formation mean direction (dec., 339.4°; inc., 19.9°, k = 32.2, α_{95} = 7.8°) is a primary magnetization direction, which was acquired during cooling of the Corral Canyon ash flow tuffs. The single red bed site (CH075) yields a mean direction located in the middle of the cluster of ash flow site directions.

Paleomagnetic Pole Position

The use of paleomagnetic poles from southeastern Arizona for analysis of North American APW has been questioned because of the tectonic history of this region [Steiner, 1983]. Kluth et al. [1982] address this question and point to a number of reasons why vertical axis rotation is unlikely. Pre-Basin and Range paleogeographic reconstructions restoring about 20% crustal extension do not require vertical axis rotation [Coney, 1978]. Various forms of structural orientation data also indicate that no significant vertical axis tectonic rotation took place in southeastern Arizona during Laramide or Basin and Range orogenies.

The Corral Canyon sequence can be characterized as a "homoclinal" domain following Davis [1979]. Poles to bedding are dominantly unimodal to the south, thus prohibiting evaluation of structural plunge. The nearest "cylindrical" fold domains as recognized by Davis [1979] are the southern Santa Rita and Huachuca mountains, where the characteristic fold axes plunge 0° and 4°, respectively. Although plunge cannot be directly evaluated from our structural data within the Corral Canyon rocks, the regional

TABLE 3. Average Paleomagnetic Data From the Corral Canyon Sequence

	N	Dec., deg.	Inc., deg.	α_{95} deg.	k
Average of site mean directions					
In situ	12	326.1	51.0	9.1	23.5
Corrected	12	339.4	19.9	7.8	32.2
	N	North Latitude, deg.	East Longitude, deg.	α_{95}	k
Average of VGP's					
In situ	12	61.2	172.6	9.9	20.3
Corrected	12	61.8	116.0	6.2	49.6

Corrected, structurally corrected for local tilt of bedding. Formation location for bottom half of table is 31.4°N, 249.3°E.

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11545 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

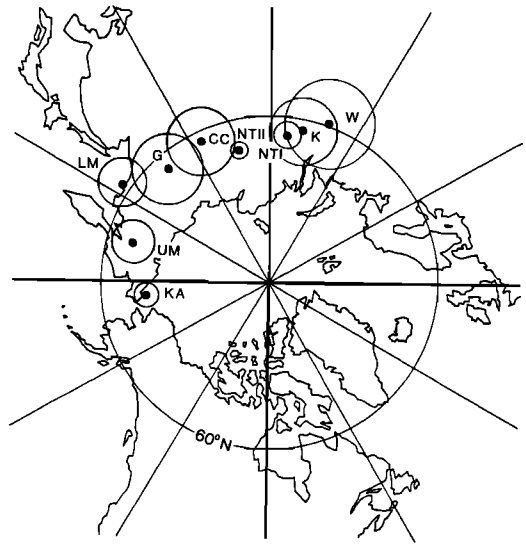


Fig. 6. Stereographic north polar projection showing the location of the Corral Canyon (CC) pole and other reliable Jurassic poles. W, Wingate Formation (200-206 Ma); K, Kayenta Formation (194-200 Ma); NTI, Newark Trend Group I (195±4 Ma); NTII, Newark Trend Group II (179±3 Ma); G, Glance Conglomerate (Canelo Hills) (151±2 Ma); LM, lower Morrison Formation (149 Ma); UM, upper Morrison Formation (145 Ma); KA, Cretaceous average of Mankinen [1978] (130-85 Ma). Mean pole locations shown with solid circle and associated A_{95} confidence region.

geology suggests that significant plunge is uncommon for Laramide folds of this area. Accordingly, we believe structural plunge to be an unlikely source of error for the location of the Corral Canyon paleomagnetic pole. Laramide deformation in southeastern Arizona resulted in a "basement cored uplift" structural style with associated homoclinal and shallowly to nonplunging cylindrical fold domains [Davis, 1979] rather than regional, thin-skinned overthrusting, which would be more likely to yield vertical axis rotations.

Rehrig and Heidrick [1976] found consistent orientations of fractures and dikes of Laramide and mid-Tertiary age across southeastern Arizona and into the transition zone bordering the Colorado Plateau. The available paleomagnetic data from southeastern Arizona also speak strongly against vertical axis tectonic rotation of the Jurassic volcanics. Paleomagnetic poles from Laramide volcanic rocks in the same area are concordant with poles of similar age from east of the Rocky Mountains in central Montana [Vugteveen et al., 1981].

The paleomagnetic pole for the Corral Canyon rocks calculated from 12 site mean VGP's is located at 61.8°N, 116.0°E, α_{95}= 6.2°, k = 49.6 (Table 3). This pole is shown in Figure 6 along with other reliable Jurassic paleomagnetic poles for cratonic North America. The Corral Canyon pole is concordant with a latitudinal arc of apparent polar wander during the 200 to 150 Ma interval. This track begins at the Wingate Formation pole [Reeve, 1975] and progresses eastward through the Kayenta Formation pole [Steiner and Helsley, 1974], the Newark Trend Group I and Group II poles [Smith and Noltimier, 1979], the Corral Canyon pole, the Glance Conglomerate (Canelo Hills) pole [Kluth et al., 1982], and the lower Morrison Formation pole [Steiner and Helsley, 1975]. The location of the Corral Canyon pole between the 151±2 Ma Glance Conglomerate pole and the 179±3 Ma Newark Group II pole is consistent with the previously discussed Rb/Sr isochron age of 172±5.8 Ma. We find the spatiotemporal consistency of the Corral Canyon pole with respect to these other poles to be quite compelling. An analysis of North American Jurassic APW is given in the accompanying paper [May and Butler, this issue].

Summary

Intermediate to silicic volcanic rocks in the Patagonia Mountains of southeastern Arizona are part of an autochthonous segment of the Jurassic Cordilleran magmatic arc. A 650-m-thick homoclinal sequence near Corral Canyon includes welded ash flow tuff, tuff breccia, lava, and fluvial red beds. Rubidium/strontium isotopic analysis of eight whole-rock tuff samples yields an isochron age of 172±5.8(2σ) Ma and an initial $^{87}Sr/^{86}Sr$ ratio of 0.7085 ± 0.0003(2σ). Paleomagnetic samples from 18 separate welded tuffs and from a single site in red mudstone reveal a primary magnetization carried in both magnetite and hematite. Characteristics of blocking temperature and coercivity spectra indicate variable degrees of deuteric oxidation of magnetite to hematite. Both magnetic phases record the same direction of magnetization, and a selected set of 12 site mean directions yields a formation mean of dec, 339.4°; inc, 19.9°; α_{95}= 7.8°. Site mean directions pass a reversals test and define a paleomagnetic pole located at 61.8°N, 116.0°E (α_{95}= 6.2°), which is consistent with reliable Middle and Late Jurassic paleopoles from cratonic North America.

Acknowledgments. We thank Steve Wust, Laurel Vedder, Chuck Kluth, Paul Ryberg, Pat Butler, Bruno Galbrun, and Gary Calderone for assistance in the field. Assistance with lab work was received from Mike Coffin and Dave Steinke. A thorough and helpful review by Jack Hillhouse led to improvements in the manuscript. This research was funded by National Science Foundation grant EAR-8314113.

References

Baker, R. C., The geology and ore deposits of the southeast portion of the Patagonia Mountains, Santa Cruz County, Arizona, Ph.D. dissertation, Univ. of Mich., Ann Arbor, 1961.

Beck, M. E., Jr., Paleomagnetic record of plate margin tectonic processes along the western edge of North America, J. Geophys. Res., 85, 7115-7131, 1980.

Bryan, P., and R. G. Gordon, Maximum likelihood estimates of the rotation of the Colorado Plateau inferred from paleomagnetic data, Eos Trans. AGU, 66 , 254, 1985.

Coney, P. J., The plate tectonic setting of southeastern Arizona, Land of Cochise, Field

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11545 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Conf. Guideb. N. M. Geol. Soc., 29, 285-290, 1978.

Coney, P. J., D. L. Jones, and J. W. H. Monger, Cordilleran suspect terranes, Nature, 288, 329-333, 1980.

Damon, P. E., M. Shafiqullah, and K. F. Clark, Age trends of igneous activity in relation to metallogenesis in the southern Cordillera, Relations of Tectonics to Ore Deposits in the Southern Cordillera, Ariz. Geol. Soc. Dig., 14, 137-154, 1981.

Davis, G. H., Laramide folding and faulting in southeastern Arizona, Am. J. Sci., 279, 543-569, 1979.

Dunlop, D. J., Magnetic mineralogy of unheated and heated red sediments by coercivity spectrum analysis, Geophys. J. R. Astron. Soc., 27, 37-55, 1972.

Fisher, R. A., Dispersion on a sphere, Proc. R. Soc. London, Ser. A, 217, 295-305, 1953.

Gordon, R. G., A. Cox, and S. O'Hare, Paleomagnetic Euler poles and the apparent polar wander and absolute motion of North America since the Carboniferous, Tectonics, 3, 499-537, 1984.

Harland, W. B., A. V. Cox, P. G. Llewellyn, C. A. G. Pickton, A. G. Smith, and R. Walters, A Geologic Time Scale, Cambridge University Press, New York, 1982.

Hayes, P. T., Cretaceous paleogeography of southeastern Arizona and adjacent areas, U.S. Geol. Surv. Prof. Pap., 658-B, 42 pp., 1970.

Hayes, P. T., and H. Drewes, Mesozoic depositional history of southeastern Arizona, Land of Cochise, Field Conf. Guideb. N. M. Geol. Soc., 29, 201-207, 1978.

Hillhouse, J. W., Paleomagnetism of the Triassic Nikolai Greenstone, McCarthy Quadrangle, Alaska, Can. J. Earth Sci., 14, 2578-2592, 1977.

Hillhouse, J. W., and C. S. Gromme, Northward displacement an accretion of Wrangellia: New paleomagnetic evidence from Alaska, J. Geophys. Res., 89, 4461-4477, 1984.

Kirschvink, J. L., The least squares line and plane and the analysis of paleomagnetic data, Geophys. J. R. Astron. Soc., 62, 699-718, 1980.

Kluth, C. F., Geology and mid-Mesozoic tectonics of the northern Canelo Hills, Santa Cruz County, Arizona, Ph.D. dissertation, Univ. of Ariz., Tucson, 1982.

Kluth, C. F., Geology of the northern Canelo Hills and implications for the Mesozoic tectonics of southeastern Arizona, in Mesozoic Paleogeography of West-Central United States, edited by M. W. Reynolds, and E. D. Dolly, pp. 159-171, Rocky Mountain Section, Society of Economic Paleontologists and Mineralogists, Denver, Colo., 1983.

Kluth, C. F., R. F. Butler, L. E. Harding, M. Shafiqullah, and P. E. Damon, Paleomagnetism of Late Jurassic rocks in the northern Canelo Hills, southeastern Arizona, J. Geophys. Res., 87, 7079-7086, 1982.

Knight, R., Structure and mineralization of the Oro Blanco mining district, Santa Cruz County, Arizona, Ph.D. dissertation, Univ. of Ariz., Tucson, 1970.

Ludwig, K. R., Plotting and regression programs for isotope geochemists, for use with HP-86/87 microcomputers, U.S. Geol. Surv. Open File Rep. 83-849, 103 pp., 1983.

Mankinen, E. A., Paleomagnetic evidence for a Late Cretaceous deformation of the Great Valley sequence, Sacramento Valley, California, J. Res. U.S. Geol. Surv., 6, 383-390, 1978.

Marvin, R. F., C. W. Naeser, and H. H. Mehnert, Tabulation of radiometric ages including unpublished K-Ar and fission-track ages for rocks in southeastern Arizona and southwestern New Mexico, Land of Cochise, Field Conf. Guideb. N. M. Geol. Soc., 29, 243-252, 1978.

May, S. R., Paleomagnetism of Jurassic volcanic rocks in southeastern Arizona and North American Jurassic apparent polar wander, Ph.D. dissertation, Univ. of Ariz., Tucson, 1985.

May, S. R., and Butler, R. F., North American Jurassic Apparent Polar Wander: Implications for plate motion, paleogeography, and Cordilleran tectonics, J. Geophys. Res., this issue.

O'Reilly, W., Rock and Mineral Magnetism, Blackie, Chapman and Hall, New York, 1984.

Reeve, S. C., Paleomagnetic studies of sedimentary rocks of Cambrian and Triassic age, Ph.D. dissertation, Univ. of Tex. at Dallas, 1975.

Rehrig, W. A., and T. L. Heidrick, Regional tectonic stress during the Laramide and late Tertiary intrusive periods, Basin and Range province, Arizona, Ariz. Geol. Soc. Dig., 10, 205-228, 1976.

Shafiqullah, M., P. E. Damon, D. J. Lynch, P. H. Kuck, and W. H. Rehrig, Mid-Tertiary magmatism in southeastern Arizona, Land of Cochise, Field Conf. Guideb. N. M. Geol. Soc., 29, 231-242, 1978.

Simons, F. S., Mesozoic stratigraphy of the Patagonia Mountains and adjoining areas, Santa Cruz County, Arizona, U.S. Geol. Surv. Prof. Pap. 658-E, 23 pp., 1972.

Simons, F. S., Geologic map and sections of the Nogales and Lochiel quadrangles, Santa Cruz County, Arizona, U.S. Geol. Surv. Misc. Invest. Ser. I-762, 1974.

Smith, T. E. and H. C. Noltimier, Paleomagnetism of the Newark trend igneous rocks of the north central Appalachians and the opening of the central Atlantic Ocean, Am. J. Sci., 279, 778-807, 1979.

Steiner, M. B., Mesozoic apparent polar wander and plate motions of North America, in Mesozoic Paleogeography of West-Central United States, edited by M. W. Reynolds, and E. D. Dolly, pp. 1-11, Rocky Mountain Section, Society of Economic Paleontologists and Mineralogists, Denver, Colo., 1983.

Steiner, M. B., and C. E. Helsley, Magnetic polarity sequence of the Upper Triassic Kayenta Formation, Geology, 2, 191-194, 1974.

Steiner, M. B., and C. E. Helsley, Reversal pattern and apparent polar wander for the Late Jurassic, Geol. Soc. Am. Bull., 36, 1537-1543, 1975.

Stoyanow, A. A., Lower Cretaceous stratigraphy in southeastern Arizona, Geol. Soc. Am. Mem., 38, 169 pp., 1949.

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11545 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Vedder, L. K., Stratigraphic relationship of Late Jurassic Canelo Hills volcanics and Glance Conglomerates, southeastern Arizona, M.S. thesis, Univ. of Ariz., Tucson, 1984.

Vugteveen, R. W., A. E. Barnes, and R. F. Butler, Paleomagnetism of the Roskruge and Gringo Gulch volcanics, southeast Arizona, J. Geophys. Res., 86, 4021-4028, 1981.

R. F. Butler, P. E. Damon, and M. Shafiqullah, Department of Geosciences, University of Arizona, Tucson, AZ 85721.

S. R. May, Exxon Production Research Company, P.O. Box 2189, Houston, TX 77252.

(Received January 16, 1986;
revised June 19, 1986;
accepted June 24, 1986.)

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11545 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License